

Jiaming Hu

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Miami, FL - 33155, United States

RESEACH INTERESTS

AI-driven Drug Discovery • Computational Chemistry • Machine Learning • Deep Learning • Generative AI • Agentic AI • Multimodal and Multi-omics Studies • Precision Oncology • Cheminformatics • Bioinformatics • Molecular Modeling

WORK EXPERIENCE

Sylvester Comprehensive Cancer Center

Feb 2026 - Present

Cheminformatics and Bioinformatics Data Scientist

Miami, FL

- Conduct research in computational drug discovery and biomedical data science using large-scale multimodal, multi-omics, and molecular profiling data
- Design, build, and deploy multi-agent AI systems with integrated predictive and generative AI for end-to-end computational-driven drug discovery, orchestrating automated workflows from data curation to lead optimization to translational modeling and precision medicine decision-making.
- Lead and contribute to peer-reviewed scientific publications, grant proposals, and interdisciplinary research collaborations

University of Miami Miller School of Medicine

Aug 2020 - Dec 2025

Graduate Researcher | Advisor: Stephan C. Schürer, Ph.D.

Miami, FL

- Develop machine learning and deep learning models for predictive and generative modeling in drug discovery and precision oncology
- Conduct large-scale multi-omics and multimodal biomedical data to study drug-target interactions and cellular responses
- Support the development of cloud-based research data platform for large-scale data management and analysis
- Collaborate with cross-functional teams to validate computational predictions and present findings through peer-reviewed publications and scientific conferences

Nucleate FL

Oct 2024 - Jun 2025

Computational Scientist

Miami, FL

- Developed machine learning models and applied molecular modeling tools in a collaborative biotech research environment
- Led large-scale data-driven projects, integrating computational methods and cross-disciplinary efforts to drive innovation and translational research

University of Miami

Jan 2019 - Jan 2021

Undergraduate Researcher | Advisor: Grace R. Zhai, Ph.D.

Miami, FL

- Conducted laboratory experiments using *Drosophila melanogaster* to assess toxin-induced changes in brain health and neurological function
- Collaborated with cross-functional teams to investigate the biological effects of environmental toxins and their potential link to neurodegenerative diseases.

EDUCATION

University of Miami

Dec 2025

Ph.D. in Biomedical Science - Human Genetics and Genomics | Advisor: Stephan C. Schürer, Ph.D.

Miami, FL

- Dissertation: Multimodal Data-Driven Computational Models for Kinase Inhibitor Discovery and Cellular Drug Response Prediction

University of Miami

May 2020

B.S. in Marine Science and Biology

Miami, FL

PROJECTS

- **Project A: Agentic Target and Lead Analysis Suite (ATLAS)** 2026-Ongoing
Tools: Python, R, KNIME, RDKit, DeepChem, DuckDB, MCP, LangChain, Azure, Claude Code
 - Lead cheminformatics strategy and platform development for ATLAS, an AI-driven oncology drug discovery platform.
 - Design and develop agentic AI systems to identify druggable targets, actionable biomarkers, and lead compounds from multimodal biological contexts.
 - Architect and deploy a multi-agent platform that orchestrates and automates the end-to-end drug discovery pipeline, from data integration and target activity to predicted selectivity, virtual screening, docking, molecular dynamics simulations, and ADMET/PK-PD modeling.
 - Implement generative AI pipelines for de novo molecular design and lead optimization to accelerate candidate discovery.
 - Collaborate with cross-functional bioinformatics, software engineering, and translational research teams to advance drug discovery.
- **Project B: Structure-Based Transformer Model for Kinase Binding Affinity Prediction** 2026-Ongoing
Tools: Python, KNIME, RDKit, DiffDock, Boltz, AlphaFold, OpenMM
 - Lead development of a structure-informed transformer model to predict kinase–ligand binding affinity across the human kinome.
 - Integrate protein sequences, 3D binding pocket features, and ligand structures for multi-modal learning.
 - Develop interpretable AI methods to identify key molecular interactions and multi-target kinase opportunities.
 - Prioritize drug-like candidates targeting understudied (“dark”) kinases for precision oncology.
- **Project C: Sylvester Data Portal - Cloud-based Multi-Omics Research Platform** 2024-2025
Tools: Python, R, KNIME, Pipeline Pilot, RDKit, TileDB, Datalake, Azure
 - Co-develop a cloud-based platform for secure storing, managing, and analyzing large-scale multi-omics data, accelerating cross-team collaboration and data accessibility.
 - Enable integration of diverse transcriptomic, proteomic, and drug response datasets to support real-time computational modeling and discovery workflows.
 - Facilitated reproducible data management practices and transparent research dissemination across internal and external collaborators.
 - Web Page:
<https://sdp.miami.edu>
- **Project D: AI-Powered Platform for Kinome-Wide Virtual Screening in Precision Oncology** 2024-2025
Tools: Python, Docker, Azure, GCP, Schrödinger Maestro, AlphaFold, ChemAxon, InfiniSee, RDKit
 - Developed the KNet platform for kinome-wide virtual screening and prediction of kinase binding affinities, enabling rapid prioritization of active compounds.
 - Applied structure-based modeling and molecular simulations to uncover binding modes and conformational dynamics of understudied kinases.
 - Collaborated with experimental teams to design and implement *in vitro* binding assays, establishing an iterative computational-experimental loop that improved prediction accuracy and accelerated hit-to-lead optimization.
 - Github Repository:
[KNet: A Web-Based Deep Learning Platform for Kinome-Wide Virtual Screening](#)
- **Project E: Pan-Cancer Multimodal Models for Drug Sensitivity Prediction** 2023-2024
Tools: Python, R, Tensorflow, KNIME, Machine Learning, Deep Learning, RDKit
 - Constructed multimodal models integrating molecular fingerprints, transcriptional consensus signatures, and multi-omics features to predict cellular drug sensitivity across cancer-types.
 - Developed scalable, automated pipelines for high-throughput model training, evaluation, and reproducibility.
 - Generated pan-cancer predictive insights linking compound response patterns to transcriptomic and proteomic profiles, advancing precision oncology research
 - Github Repository:
[Pan-Cancer Drug SensitivitySeq](#)
- **Project F: Kinome-Wide Deep Learning Framework for Biochemical Activity Prediction** 2022 - 2024
Tools: Python, Pytorch, Tensorflow, SQL, Machine Learning, Deep Learning, DeepChem, Scikit-learn
 - Developed a multi-task deep learning framework to predict ligand-target biochemical activity across the kinome, enabling large-scale virtual screening and mechanism-of-action analysis
 - Integrated diverse chemical and structural data sources to build a unified foundation for structure-based drug discovery and model generalization across novel chemotypes.
 - Enhanced biomedical ontologies (BAO, DTO) to support standardized data integration and interoperable modeling pipelines for predictive toxicology and pharmacology.
 - Github Repository:
[Kinome-wide-Virtual-Screening-by-Multi-task-Deep-Learning](#)

PUBLICATIONS AND PREPRINTS

P=PUBLISHED, S=IN SUBMISSION, C=CONFERENCE, T=THESIS

- [P.1] Glenny-Pescov, J., Chung, C., Ross, N., **Hu, J.**, Sinclair, M., Khurshid, R., Karlsson, A., Schürer, S.C. (2026). **Advancing the bioassay ontology through integrated PK/PD and safety pharmacology representation.** Journal of Biomedical Semantics, 17, 6. <https://doi.org/10.1186/s13326-025-00342-5> [Impact Factor: 2.0]
- [T.1] **Hu J.** (2025). Multimodal Data-Driven Computational Models for Kinase Inhibitor Discovery and Cellular Drug Response Prediction. PhD Thesis, University of Miami.
- [S.1] **Hu, J.**, Rupprecht, L., Schürer, S.C. (2025). KNet: A Web Application for Kinome-Wide Small Molecules Drug Discovery in Cancer. Manuscript under review for publication in *iScience*. Preprint available at [SSRN eLibrary](#)
- [S.2] Ocasio, B.A., **Hu, J.**, Stathias, V., Martinez, M.J., Burnstein, K.L., Schürer S.C. (2024). Pan-Cancer Drug Sensitivity Prediction from Gene Expression using Deep Learning. Prepare for submission. Preprint available at [bioRxiv](#)
- [C.1] Ocasio, B.A., **Hu, J.**, Stathias, V., Martinez, M.J., Burnstein, K.L., Schürer S.C. (2024). **Pre-clinical pan-cancer drug repurposing via deep learning.** Abstract presented at the European Society for Medical Oncology (ESMO) Congress, published in ESMO Open. [Impact Factor: 8.3]
- [P.2] **Hu, J.**, Allen, B.K., Stathias, V., Ayad, N.G., Schürer, S.C. (2024). **Kinome-Wide Virtual Screening by Multi-Task Deep Learning.** International Journal of Molecular Sciences, 25(5):2538. <https://doi.org/10.3390/ijms25052538> [Impact Factor: 4.9]
- [P.3] Khurshid, R., Schulz, J. M., **Hu, J.**, Snowden, T. S., Reynolds, R. C., Schürer, S. C. (2024). **Targeted degrader technologies as prospective SARS-CoV-2 therapies.** Drug discovery today, 29(1), 103847. <https://doi.org/10.1016/j.drudis.2023.103847> [Impact Factor: 7.5]
- [P.4] **Hu, J.**, Liu, J., Zhu, Y., Diaz-Perez, Z., Sheridan, M., Royer, H., Leibensperger, R., 3rd, Maizel, D., Brand, L., Pendorf, K. J., Gaston, C. J., Zhai, R. G. (2020). **Exposure to Aerosolized Algal Toxins in South Florida Increases Short- and Long-Term Health Risk in Drosophila Model of Aging.** Toxins, 12(12), 787. <https://doi.org/10.3390/toxins12120787> [Impact Factor: 4.0]

PRESENTATIONS AND POSTERS

O=ORAL, P=POSTER

- [P.1] Kinome-Wide Virtual Screening by Multi-task Deep Learning for Small Molecule Drug Discovery in Cancer. 24th Annual Zubrod Memorial Lecture and Cancer Research. May 2025. Poster
- [P.2] Kinome-Wide Virtual Screening by Multi-task Deep Learning for Small Molecule Drug Discovery in Cancer. Drug Discovery Chemistry. Apr 2025. Poster
- [O.1] Pre-clinical pan-cancer drug repurposing via deep learning. Molecular Analysis for Precision Oncology Congress (MAP). Oct 2024. Oral Presentation.
- [P.3] Predict and Prioritize Small Molecule Inhibitors against PNCK using Kinome-wide Multi-task Deep Neural Network Classifiers. Inaugural Computing Day Poster Presentation. Apr 2023. Poster
- [O.2] Structural-based Molecular Simulation and Machine Learning Studies to Identify Inhibitors Against SARS-CoV-2 Papain-like Protease (PLpro). 22nd Annual Zubrod Memorial Lecture and 3-minute Virtual Cancer Research Trainee Talks. May 2021. Oral Presentation.
- [P.4] Exposure to Aerosolized Microcystin of Harmful Algal Blooms in Lake Okeechobee Poses Negative Impact on Health in a Drosophila Model of Aging. Summer Undergraduate Research Fellowship (SURF) Symposium. Aug 2019. Poster.

TEACHING EXPERIENCE

Teaching Assistant – HGG630 Variation and Disease

University of Miami Miller School of Medicine | Professor: Holly N. Cukier, Ph.D.

Spring 2024

Miami, FL

Tutor – Camner Center for Academic Resources

University of Miami

Sep 2019 – Mar 2020

Coral Gables, FL

CLINICAL TRAINING

Human Genetics Clinical Rotation

University of Miami Miller School of Medicine

Fall 2023

Miami, FL

SKILLS

- **Programming:** Python, R, SQL, Git, Bash, Docker, Claude Code
- **Computing Environments:** Linux/Unix systems, HPC clusters, shell scripting, command-line tools
- **Web App Development:** Streamlit, React, FastAPI
- **ML/DL:** Machine learning, Deep learning, Generative AI, Tensorflow, PyTorch, DeepChem, Scikit-learn
- **Computational Biology/Chemistry Tools:** Schrödinger Maestro Suite - Molecular Docking, Molecular Dynamics Simulation, AutoDock Vina, AlphaFold, Boltz, OpenMM, GROMACS, PyMOL, ChemAxon, InstantJchem, BioSolveIT InfiniSee, RDKit
- **Data Analysis:** KNIME, Pipeline Pilot, Statistical Analysis
- **Data Visualization:** Tableau, Spotfire, Matplotlib, Seaborn
- **Cloud Computing:** Microsoft Azure, Google Cloud Platform

HONORS AND AWARDS

- **Award of Academic Merit** Dec 2025
University of Miami
- **Graduate Student Travel Award** April 2025
University of Miami
- **U-LINK fellowship** 2019-2021
University of Miami
- **Provost's Honor Roll** Spring 2018, Fall 2019
University of Miami
- **Dean's List** Spring 2018, Spring 2019, Fall 2019
University of Miami
- **National Honor Society** 2016

EXTERNAL RESEARCH PROJECTS

- **Project A: S2P Dynamics** Feb 2026-Ongoing
AI-Driven neurotherapeutic Drug Discovery Startup
 - Designed and trained graph-based and transformer-based models collaboratively for prediction of blood–brain barrier permeability, CNS exposure, and ADMET properties to support early-stage neurotherapeutic drug discovery.
 - Contributed to the development of a cloud-based AI platform enabling rapid in silico screening and prioritization of drug candidates before synthesis.
- **Project B: Merck Challenge 2026** July 2026
In Silico Innovation Cup for AI-Agents
 - Architected a multi-agent AI system to automate hypothesis generation, literature review, model development, and decision-making in a drug discovery challenge setting.
 - Contributed to the generation of publication-ready scientific reports by orchestrating agent collaboration, result validation, and automated manuscript drafting.
 - Awarded first place in the *In Silico Innovation Cup for AI Agents*, selected by expert juries for innovation and impact.

PROFESSIONAL MEMBERSHIPS

- **American Association for Cancer Research (AACR)**
- **Sylvester Comprehensive Cancer Center Trainee Member (SCCC)**

CERTIFICATIONS

- **Academy Accreditation – Generative AI Fundamentals** Issued Jan 2025 · Expires Jan 2027
Databricks
- **Azure Certification** Issued Jul 2024
Great Learning

ADDITIONAL INFORMATION

- **Languages:** English (Full professional proficiency), Mandarin (Native)
- **Soft Skills:** Teamwork, Collaboration, Problem-solving, Time-management, Multi-tasking, Communication, Scientific writing and presentation skills
- **Interests:** Hiking, Tennis, Art museums, Films, Travel

REFERENCES

1. **Stephan C. Schürer**

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Director of Digital Drug Discovery, Institute for Data Science & Computing
Associate Director of Data Science, Sylvester Comprehensive Cancer Center
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Relationship: Advisor

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3. **Nicolette C. Ross**

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